

A No-Wait Single-Intersection Scheduler by Priority Enumeration: A Decision Tree with a Longest-Path Timing Model

1 Problem Formulation

Let $\mathcal{N} = \{1, \dots, N\}$ index the vehicles to be scheduled through the merging zone, and let $\mathcal{M}$ be the set of conflict points (*spaces*). Vehicle n traverses an ordered sequence of sub-tasks $s \in \mathcal{S}_n = \{1, \dots, S_n\}$; sub-task (n, s) occupies conflict point $\sigma(n, s) \in \mathcal{M}$ for a fixed duration $c_{n,s} > 0$. Define the prefix offset and the total traversal length

$$\pi_{n,s} = \sum_{k=1}^{s-1} c_{n,k}, \quad C_n = \sum_{s \in \mathcal{S}_n} c_{n,s}, \quad (1)$$

and let α_n denote the earliest admissible entry (release) time of vehicle n .

1.1 No-wait motion

Each vehicle crosses the intersection without stopping, so its sub-tasks are contiguous. The schedule of vehicle n is therefore parameterized by a *single* entry time $t_n \geq \alpha_n$, under which sub-task (n, s) occupies $\sigma(n, s)$ over the interval

$$I_{n,s}(t_n) = [t_n + \pi_{n,s}, t_n + \pi_{n,s} + c_{n,s}]. \quad (2)$$

1.2 Scheduling problem

Two sub-tasks conflict iff they use the same conflict point; collect such pairs in

$$\mathcal{K} = \{ \{ (n, s), (n', s') \} : \sigma(n, s) = \sigma(n', s'), n \neq n' \}. \quad (3)$$

A schedule is *feasible* iff conflicting occupations never overlap. Minimizing the total delay yields

$$\begin{aligned} \min_{t \in \mathbb{R}^N} \quad & \sum_{n \in \mathcal{N}} (t_n - \alpha_n) \\ \text{s.t.} \quad & t_n \geq \alpha_n, \quad n \in \mathcal{N}, \\ & I_{n,s}(t_n) \cap I_{n',s'}(t_{n'}) = \emptyset, \quad \forall \{ (n, s), (n', s') \} \in \mathcal{K}. \end{aligned} \quad (\text{P})$$

Since motion is no-wait, the exit time of n is $t_n + C_n$, so $\sum_n (t_n - \alpha_n)$ equals the total completion delay.

2 Disjunctive Form and the Selection

Each non-overlap constraint is a disjunction. For $\{(n, s), (n', s')\} \in \mathcal{K}$, the orientation “ (n, s) before (n', s') ” reads

$$t_{n'} - t_n \geq w_{n \rightarrow n'} := \pi_{n,s} + c_{n,s} - \pi_{n',s'}, \quad (4)$$

and the reverse orientation has weight $w_{n' \rightarrow n} = \pi_{n',s'} + c_{n',s'} - \pi_{n,s}$. A *selection* ξ orients every conflict pair, i.e. it is a set of weighted precedence arcs $i \xrightarrow{w} j$ on the vehicle nodes. Adjoining the release arcs $0 \xrightarrow{\alpha_n} n$ from a source node 0 yields the precedence graph $G_\xi = (\{0\} \cup \mathcal{N}, A_\xi)$.

3 Timing Model

Proposition 1 (Earliest-entry timing). *If G_ξ is acyclic, then problem (P) restricted to the orientation ξ has a unique pointwise-minimal solution $t^*(\xi)$, characterized by the longest-path recursion*

$$t_n^*(\xi) = \max\left(\alpha_n, \max_{i: (i \rightarrow n) \in A_\xi} (t_i^*(\xi) + w_{i \rightarrow n})\right), \quad n \in \mathcal{N}, \quad (5)$$

computed by the fixed-point (Bellman–Ford) iteration $\text{EarliestT}(\xi)$. If G_ξ contains a positive cycle, ξ is infeasible.

Equation (5) is the *timing model* $\text{EarliestT} : \xi \mapsto t^*(\xi)$: it maps an ordering to the unique contention-free, no-wait schedule that starts every vehicle as early as the chosen precedences permit. Its cost is

$$J(\xi) = \sum_{n \in \mathcal{N}} (t_n^*(\xi) - \alpha_n). \quad (6)$$

4 Reduction to a Finite Search

Proposition 2. $\min (P) = \min_{\xi \in \Xi} J(\xi)$, where Ξ is the finite set of acyclic selections.

The optimum is attained by a semi-active schedule; hence it suffices to search over orderings, the timing model (5) supplying the optimal entry times of each.

5 Decision-Tree Search

A node of the search tree is a *partial* selection $\hat{\xi}$ with decided set $D \subseteq \mathcal{K}$. The search interleaves the timing model (5) with conflict detection. The conflict oracle $\text{FindConflict}(t, D)$ returns the first pair $\{(n, s), (n', s')\} \in \mathcal{K} \setminus D$ whose intervals overlap at the current schedule t , i.e. satisfying

$$t_n + \pi_{n,s} < t_{n'} + \pi_{n',s'} + c_{n',s'} \quad \wedge \quad t_{n'} + \pi_{n',s'} < t_n + \pi_{n,s} + c_{n,s}, \quad (7)$$

and NONE if no such pair exists.

The set of leaves enumerates $\{t^*(\xi) : \xi \in \Xi\}$, i.e. all contention-free complete schedules; the minimum-cost leaf solves (P). A positive-cycle test in EarliestT prunes infeasible orientations, so only acyclic selections are expanded.

Algorithm 1 No-wait enumeration scheduler

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1: procedure RECURSE( $\hat{\xi}, D$ )
2:    $t \leftarrow \text{EarliestT}(\hat{\xi})$  ▷ timing model, Prop. 1
3:   if  $t = \text{NONE}$  then
4:     return ▷ positive cycle  $\Rightarrow$  infeasible (prune)
5:   end if
6:    $\kappa \leftarrow \text{FindConflict}(t, D)$  ▷ first overlapping, undecided pair (7)
7:   if  $\kappa = \text{NONE}$  then
8:     record leaf  $t^* \leftarrow t$ ; return ▷ conflict-free complete schedule
9:   end if
10:   $\{(n, s), (n', s')\} \leftarrow \kappa$ 
11:  RECURSE( $\hat{\xi} \cup \{n \xrightarrow{w_{n \rightarrow n'}} n'\}, D \cup \{\kappa\}$ ) ▷  $(n, s)$  before  $(n', s')$ 
12:  RECURSE( $\hat{\xi} \cup \{n' \xrightarrow{w_{n' \rightarrow n}} n\}, D \cup \{\kappa\}$ ) ▷  $(n', s')$  before  $(n, s)$ 
13: end procedure
14: Initialize: RECURSE( $\emptyset, \emptyset$ )

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